Hall Ticket Number:

MR-22

Code: B424CM5S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, December 2024/January 2025

Object Oriented Programming through Java

(Computer Science and Engineering (AI and ML))

Time: 3 Hours | *Answer ALL Questions* | Max. Marks: 60

Part-A (10 x 1 = 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	What is the significance of the 'this' keyword in Java?	1	2	1	3
1. b)	What are Java buzzwords? List any two.	1	1	1	1
1. c)	Explain the term polymorphism.	1	1	2	1
1. d)	What is the purpose of the 'super' keyword in Java?	1	2	2	2
1. e)	Define the term "multithreading".	1	1	3	2
1. f)	What is the use of the 'try' keyword in exception handling?	1	2	3	3
1. g)	State the purpose of checkbox group AWT component.	1	1	4	3
1. h)	What is the difference between a button and a canvas in AWT?	1	2	4	3
1. i)	List differences between applets and applications.	1	1	5	1
1. j)	Define MVC architecture in the context of Swing.	1	2	5	3

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Explain the concept of constructors in Java with examples.	5	2	1	2
2. b)	Describe the life cycle of a Java object and the role of garbage collection.	5	1	1	3
	OR				
2. c)	Develop a Java program to demonstrate the use of nested and inner classes.	5	1	1	3
2. d)	Explain the various types of operators in Java with examples.	5	1	1	1
3. a)	Explain hierarchical abstractions with an example.	5	2	2	2
3. b)	Develop a Java program to implement polymorphism using method overriding.	5	4	2	4
	OR				
3. c)	Explain how Java handles multiple inheritance using interfaces.	5	4	2	3
3. d)	Develop a Java program to read from and write to a file using java.io package.	5	2	2	4
4. a)	Develop a Java program to create and run multiple threads.	5	4	3	4
4. b)	Explain how to use annotations in Java with a program example.	5	3	3	3
	OR				
4. c)	Discuss the concept of synchronized threads with a program example.	5	2	3	1
4. d)	Write a Java program to create custom exception subclasses.	5	3	3	2
5. a)	Develop a Java program to handle a keyboard event using a KeyListener.	5	4	4	4
5. b)	Explain the AWT class hierarchy with an example.	5	2	4	2
	OR				
5. c)	Develop a Java Swing application that includes a tabbed pane, combo boxes and radio buttons to create a complex user interface.	5	4	4	4
5. d)	Develop a Java program to create a menu bar with menus and menu items.	5	3	4	4

P.T.O.

6. a)	Develop a Java applet to accept and display user input through parameters.	5	4	5	4
6. b)	Explain the concept and use of JComboBox in Swing with an example program.	5	2	5	2
	OR				
6. c)	Describe the differences between applets and applications in Java with examples.	5	2	5	2
6. d)	Analyze the use of JScrollPane in Swing and develop a Java program demonstrating its functionality.	5	4	5	4

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome